ENGINEERING GUIDE · YIT-DEV-DS-001 · REVISION 1.0

Device set and installation guide

Development environment for the `yit2022/yit2022` repository: the machines, the software on them, and the installation and verification procedure.

CONTENTS

1 Purpose and scope
2 Device set
3 Software set
4 Repository map
5 Installation
6 Verification
7 Operating rules
8 Troubleshooting
9 Document control

REPOSITORY
github.com/yit2022/yit2022

VALIDATED BRANCH
main @ 93c435c

DOCUMENT DATE
5 October 2026

APPLIES TO
Local checkout and Cursor Cloud Agent

INSTALL RESULT
Idempotent. No services to start.

STATUS
Validated on Ubuntu 24.04.4 LTS

This document describes how to prepare a machine to work on this repository. It is the installation guide for the repository that exists today: a GitHub profile README, with no application runtime.

1. Purpose and scope

This guide defines the device set used to read, edit, and verify `github.com/yit2022/yit2022`, and it gives the exact installation procedure for that set. Follow it when setting up a workstation or when confirming that a Cursor Cloud Agent environment can check the repository out and leave it ready to edit.

The repository contains one tracked file, `README.md`. That file is the GitHub profile page for @yit2022. It states an interest in robotic process automation and UiPath. The repository does not contain a UiPath project, a package manifest, a Dockerfile, tests, or a server. Installation therefore does not download packages and does not start a process.

1.1 What this guide covers

- The two devices in the set, and which one you need.
- Required software, with the versions observed on the validation virtual machine.
- Software that is present on the Cloud Agent image and is not used by this repository.
- The install command, how to run it twice, and the acceptance check.
- Failures that look like environment problems and are actually a wrong working directory.

1.2 What this guide does not cover

- Installing UiPath Studio, Robot, or Orchestrator. Nothing in this repository calls those products.
- Publishing the profile, GitHub account settings, or DNS.
- Production deployment. There is no deployed service.

1.3 How to read the tables

Device and software rows use stable identifiers (`DS-WS`, `SW-GIT`). “Required” means the installation in this guide fails or cannot be verified without that item. “Image only” means the tool was present on the validated Cloud Agent virtual machine and is not invoked by the install or the acceptance check.

2. Device set

A device set is the collection of machines that can perform the work. This repository has two roles. You need one of them. You do not need both.

ID	Device	Role	When to use it
`DS-WS`	Engineer workstation	Clone the repository, edit `README.md`, and run the acceptance check.	Day-to-day edits on your own computer.
`DS-CA`	Cursor Cloud Agent virtual machine	Check the repository out under `/workspace`, run the install command, and leave a shell ready for an agent.	Cloud Agent sessions, including environment builds.

Table 1. Devices in the set. Either device can complete the work.

1. CHECKOUT
Git clone or Cloud Agent checkout of `main` into the working directory.

→

2. INSTALL
`test -f README.md`, then print the ready line. The command exits.

→

3. VERIFY
Python reads the profile text and prints `HELLO_WORLD_OK`.

→

4. READY
Edit `README.md`. No daemon is left running.

2.1 Workstation requirements (DS-WS)

Any current desktop or laptop is sufficient. The procedure was not tied to a CPU count or a memory size. Provide:

- An operating system that can run Git and Bash, or Git Bash on Windows.
- Git 2.30 or newer. The validation machine used Git 2.43.0.
- Python 3.9 or newer, used only by the acceptance check in section 6. The install command itself is shell.
- A text editor. No IDE is required.
- Network access to `github.com` for the initial clone.

Disk use after clone is a few kilobytes for the Git history plus the 434-byte README. Reserve a normal working copy; there is no build output.

2.2 Cloud Agent virtual machine (DS-CA)

The procedure in section 5 was executed on a Cloud Agent virtual machine with the characteristics below. Treat the CPU and memory figures as observed capacity, not as a minimum. The install command does not allocate memory beyond a shell and, during verification, a short Python process.

Attribute	Validated value
Operating system	Ubuntu 24.04.4 LTS (Noble Numbat)
Architecture	x86_64
Processors observed	4
Memory observed	15 GiB reported by `free -h`
Login user	`ubuntu`
Workspace	`/workspace`
Shell for install	`bash -lc` (login, non-interactive)
Egress on the validation run	Unrestricted. The install command does not download anything.

Table 2. Host specification observed while validating revision 1.0.

> **Account.** Cloud Agent commands run as `ubuntu`. Do not add a second user. The install command does not use `sudo` and does not write outside the repository.

3. Software set

Install only what the device role requires. Extra language toolchains on the Cloud Agent image are listed so an agent does not mistake them for project dependencies.

3.1 Required software

ID	Component	Minimum	Validated	Used for
`SW-GIT`	Git	2.30	2.43.0 at `/usr/bin/git`	Clone and status
`SW-BASH`	Bash	4	5.2.21	Install command
`SW-PY`	Python 3	3.9	3.12.3 at `/usr/bin/python3`	Section 6 acceptance check only
`SW-CORE`	`test` , POSIX utilities	From the shell	Ubuntu coreutils	`test -f README.md`

Table 3. Software the procedure actually runs.

On Ubuntu, Git and Python 3 are installed with the distribution packages if they are missing:

```
sudo apt-get update
sudo apt-get install -y --no-install-recommends git python3 ca-certificates
```

Confirm the commands resolve before cloning:

```
git --version
python3 --version
bash --version
```

3.2 Present on the validated image, not used by this repository

The Cloud Agent base image includes compilers and runtimes this repository never calls. Do not add them to the install command. Do not remove them as part of installing this repository; they belong to the base image, not to the project.

ID	Component	Version observed	How it resolved
`SW-NODE`	Node.js on `PATH`	v22.14.0	`/exec-daemon/node`
`SW-NPM`	npm	10.9.7	nvm Node v22.22.2, `~/.nvm/versions/node/v22.22.2/bin/npm`
`SW-PNPM`	pnpm	10.33.3	Same nvm Node v22.22.2 prefix
`SW-GO`	Go	1.22.2 linux/amd64	`/usr/bin/go`
`SW-RUST`	rustc	1.83.0 (90b35a623, 2024-11-26)	`/usr/local/cargo/bin/rustc`
`SW-JDK`	OpenJDK	21.0.10+7 (Ubuntu 124.04)	`/usr/bin/java`

Table 4. Image software recorded on 5 October 2026. None of these are project dependencies.

> **Node and npm are not one install.** On the validation virtual machine, `node` was v22.14.0 from `/exec-daemon/node`, while `npm` and `pnpm` belonged to nvm's Node v22.22.2. This repository does not run Node. If a later change adds a Node project, pin one Node binary and invoke that install's `npm`, and put the install in the environment install command. Do not assume `node` and `npm` on `PATH` share a prefix.

3.3 Secrets and network

No API token, password, or private registry is required to install or verify this repository. Cloning over HTTPS uses whatever GitHub authentication the operator already has. The install command makes no outbound request. A machine that cannot reach the public internet can still run install and verification on an existing checkout.

4. Repository map

Validated against `main` at commit `93c435c1074bab627843b33e276b8e67abeb40bf` ("Create README.md", 23 May 2022) before this guide was added. After this guide is merged, the tree also contains this document and its HTML source. No other product files are part of the device set.

Path	Role
`README.md`	GitHub profile page. 434 bytes, 10 lines. The install command checks that this file exists. The acceptance check reads its text.
`docs/device-set-and-installation-guide.html`	Source for this guide.
`docs/device-set-and-installation-guide.pdf`	This guide, rendered for reading and printing.
`scripts/render-device-guide.sh`	Reprints the PDF from the HTML source with headless Chrome.

Table 5. Files an operator should expect after this guide is added. The install command still keys off `README.md` only.

The profile text that must remain readable, ignoring the HTML comment GitHub adds to profile repositories, is:

- Hi, I'm @yit2022
- I'm interested in Robotic Process automation
- I'm currently learning UIPATH
- I'm looking to collaborate on robotic process automations projects
- How to reach me rofric2020@gmail.com

The acceptance check requires the strings `yit2022`, `UIPATH`, and `rofric2020@gmail.com`, at least four visible lines, and an even number of Markdown fence markers (there are none today). Updating the biography is fine. If those strings are removed, update section 6 in the same change so the check matches the new profile.

Working directory. Every command in sections 5 and 6 assumes the current directory is the repository root, the directory that contains `README.md`. On a Cloud Agent that directory is `/workspace`.

5. Installation

Installation is intentionally small. It proves the checkout is the repository root and then exits. It does not modify files, install packages, or listen on a port. Running it again is the idempotence check: the second run must also exit 0 and print the same line.

5.1 Install command

Save this as the Cloud Agent install command, or run it from the repository root in any Bourne-compatible login shell. Keep the three lines. Do not join them with &&.

```
set -euo pipefail
test -f README.md
echo "No package dependencies. Profile README repository is ready."
```

Expected standard output, and nothing on standard error:

```
No package dependencies. Profile README repository is ready.
```

Exit status `0`. If `README.md` is absent, `test` returns status 1 and `set -e` stops the script before the echo.

5.2 Install on a workstation (DS-WS)

1 Install Git and Python 3

Use section 3.1. Skip this step when `git --version` and `python3 --version` already succeed.

2 Clone the repository

Pick an empty parent directory. The clone creates `yit2022`.

```
git clone https://github.com/yit2022/yit2022.git
cd yit2022
git status
git rev-parse --abbrev-ref HEAD
```

`git status` should report a clean work tree. The branch should be `main` unless you checked out another branch on purpose.

3 Run the install command

From the clone root:

```
bash -lc 'set -euo pipefail
test -f README.md
echo "No package dependencies. Profile README repository is ready."'
```

4 Run it a second time

Repeat the same command. A second success is the idempotence evidence. The script creates no cache, lock, or virtual environment, so the second run cannot drift from the first.

5 Run the acceptance check

Continue with section 6 before treating the machine as ready.

5.3 Install on a Cloud Agent (DS-CA)

Cursor checks the repository out into `/workspace` and then runs the install command from that directory. Put only the three lines from section 5.1 in the environment install field. Leave the start field empty.

Why there is no start command. Start is for processes that must keep running, such as a database or a development server. This repository has neither. An empty start field means no per-boot process is launched for the project. Do not put the install command in start, and do not put a server in install.

1 Confirm the workspace

`pwd` prints `/workspace`. `ls README.md` prints `README.md`. If the file is missing, stop. Do not create a substitute README to force install to pass.

2 Run install the way the environment runs it

```
bash -lc 'set -euo pipefail
test -f README.md
echo "No package dependencies. Profile README repository is ready."'
```

3 Repeat the command

The second exit status must be 0. On 5 October 2026 both runs printed the ready line and exited 0.

4 Do not start a server

There is no port to open. Readiness is the acceptance check in section 6, not an HTTP response.

5.4 What a successful install changes

Location	Change
Repository files	None. `git status --short` stays empty.
Packages	None. No `apt`, `pip`, or `npm` invocation.
Processes	None left running. The shell exits.
Standard output	One ready line.

Table 6. Side effects of the install command.

6. Verification

Run this from the repository root after install. It is the acceptance check for both devices. Python's standard library is enough; do not `pip install` anything for it.

```
python3 - <<'PY'
from pathlib import Path
text = Path("README.md").read_text(encoding="utf-8")
visible = []
for line in text.splitlines():
    stripped = line.strip()
    if not stripped or stripped.startswith("<!") or stripped.startswith("yit2022/") or
stripped.startswith("You can click") or stripped.startswith("--->"):
        continue
    visible.append(stripped)
if "```" in text and text.count("```") % 2 != 0:
    raise SystemExit("unbalanced markdown fences")
required = ["yit2022", "UIPATH", "rofric2020@gmail.com"]
missing = [item for item in required if item not in text]
if missing:
    raise SystemExit(f"profile content missing: {missing}")
if len(visible) < 4:
    raise SystemExit("profile README has too few visible lines")
print(f"bytes={len(text.encode())} lines={len(text.splitlines())} visible_lines={len(visible)}")
print("--- visible profile ---")
print("\n".join(visible))
print("HELLO_WORLD_OK")
PY
```

On the validated README the check printed:

```
bytes=434 lines=10 visible_lines=5
--- visible profile ---
- 👋 Hi, I'm @yit2022
- 👀 I'm interested in Robotic Process automation
- 🌱 I'm currently learning UIPATH
- 💞️ I'm looking to collaborate on robotic process automations projects
- 📫 How to reach me rofric2020@gmail.com
HELLO_WORLD_OK
```

> **Byte count.** `bytes=434` and `lines=10` describe the README before this guide was added. After any edit to `README.md`, those two numbers change. `HELLO_WORLD_OK` is the pass signal. A missing required string, fewer than four visible lines, or an unbalanced fence is a failure.

6.1 Acceptance checklist

The device is ready when every row is true.

#	Check	Pass condition
1	Repository root	`test -f README.md` exits 0
2	First install	Exit 0 and the ready line
3	Second install	Same result as the first install
4	Work tree	`git status --short` prints nothing
5	Tracked product file	`git ls-files` includes `README.md`
6	Acceptance check	Output ends with `HELLO_WORLD_OK`
7	Services	No project server was started

Table 7. Acceptance checklist used on 5 October 2026. Rows 1 through 6 passed on the validation virtual machine, and row 7 held because no start command is configured.

6.2 Recorded validation

Revision 1.0 of this guide records two runs of the install command on the virtual machine in Table 2, followed by the acceptance check. Both install runs printed the ready line. The acceptance check printed `HELLO_WORLD_OK` with five visible profile lines. `git status` was clean and the only tracked file at that time was `README.md`.

The same install command was then executed by a Cloud Agent environment build. The build log contains the ready line and install exit code 0. A new agent booted from build `bld-20261005-61f28ef8-c54b-4398-985a-1bddc6bd255c` reported Git 2.43.0, Python 3.12.3, and Node v22.14.0, a clean `main` checkout, no start-script directory, no tmux session, and `HELLO_WORLD_OK`.

7. Operating rules

7.1 Where new setup belongs

If you add…	Put it in…	Do not put it in…
A package install tied to a lockfile	The environment install command, after `test -f README.md`	A per-boot start command
A long-running server or database	A start command that skips work when the service is already up	The install command. Install must exit.
A one-off editor plugin on your laptop	The workstation only	The repository install command

Table 8. Where future setup belongs. None of these additions exist in the repository today.

7.2 Editing the profile

1 Edit README.md on a branch

Keep the profile readable as a short list. The HTML comment block at the bottom is a GitHub profile hint; leaving it in place does not affect the visible page.

2 Re-run section 6

If you rename the GitHub login, change the learning line so it no longer contains `UIPATH`, or replace the contact address, update the `required` list in the acceptance check in the same change.

3 Re-run install

Install still only checks that the file exists. Run it once after the edit to confirm the working directory is still the root.

7.3 Reprinting this PDF

Edit `docs/device-set-and-installation-guide.html`, then from the repository root:

```
bash scripts/render-device-guide.sh
```

The script needs Google Chrome on PATH. It writes `docs/device-set-and-installation-guide.pdf` next to the HTML. Commit the HTML and the PDF together so the printed guide matches the source.

8. Troubleshooting

Symptom	Likely cause	What to do
Install exits 1 with no ready line	`README.md` is not in the current directory.	Run `pwd` and `ls`. `cd` to the clone root, or to `/workspace` on a Cloud Agent, and run install again.
`bash: set: pipefail` or `set: -u` is rejected	The script was executed by `sh`, not Bash.	Run it with `bash -lc`, as in section 5.2.
Ready line prints, then `git status` shows modifications	Something other than this install command changed files. The install command does not write.	Inspect `git diff`. Do not attribute those edits to install.
Acceptance check prints `profile content missing`	The README no longer contains one of the three required strings.	Restore the string or update the check and this guide together.

Symptom	Likely cause	What to do
`python3: command not found`	Python 3 is not installed. Install can still succeed, because it is shell-only.	Install Python 3 (section 3.1) and re-run section 6. Do not mark the device ready on the install line alone.
Clone asks for credentials or returns 403	GitHub authentication for a private fetch, or no network path to `github.com`.	This repository's public README clone does not need a token. Confirm the URL is `https://github.com/yit2022/yit2022.git` and that HTTPS to `github.com` works.
`node -v` and `npm -v` disagree about the major install	Known image layout in Table 4. Two Node installs are on the machine.	Ignore it for this repository. If you add Node later, select one prefix deliberately.
A port is already in use during setup	Another project's server. This guide starts none.	Do not add a server to make this repository "look" ready. Readiness is `HELLO_WORLD_OK`.

Table 9. Failures and the response that keeps the device set accurate.

8.1 Commands that collect a useful report

If install or the acceptance check fails, capture this report before changing the machine. It is enough to see whether the failure is the working directory, the README, or a missing interpreter.

```
pwd
ls -la README.md
git status --short
git rev-parse --abbrev-ref HEAD
git --version
python3 --version
bash -lc 'set -euo pipefail
test -f README.md
echo "No package dependencies. Profile README repository is ready."'
```

9. Document control

Revision	Date	Description
1.0	5 October 2026	Initial device set and installation guide. Records the workstation and Cloud Agent roles, the shell install command, the acceptance check, and the validation virtual machine's software inventory.

Table 10. Revision history.

Document ID YIT-DEV-DS-001. The identifiers in the tables are local to this guide. They are not product SKUs. Versions in Table 2 and Table 4 were read from the validation virtual machine on the document date; a later base image can report different preinstalled compilers without invalidating the install command.